Date ..../..../.... No ....

Cartilage: → From Supportive Connective Tissues
→ Consist of Cells, Fibers and ground substances.

→ There's 3 Types of Cartilages: fibro Cartilage, Hyaline Cartilage and Elastic Cartilages.

① Fibro Cartilage:-

- intermediate in character between dense Connective Tissue & Cartilage.
- Have Type II Collagen. + Type I → Chondrocytes حوالين

زي ال dense C.T. عبارة عن Collagen bundle لها ثخين وبينها ...
بعضها جنبها تتابعت ب ال Cartilage الي بتكون ال Chondrocyte ها
مميزتها إنها قاعدة جوا أوضة اسمها LaCuna

- There is No Perichondrium → غلاف بيحيط بالخلية

- found in intervertebral discs or stifle joints

---

② Hyaline Cartilage:- «Most Common Type in adult + fetus»

- Have Type II Collagen only.
- Ground Substance: GAGs.
- Chondrocytes are found in Matrix but found as cell nest inside LaCuna May be 2 cells or More (2:8)
- All Hyaline Cartilage Have Perichondrium Except articular Surface of Cartilage.

→ Perichondrium Consists of 2 layers: outer fibrous layer which is dense irregular C.T. + inner Cellular layer which Contain the chondroblasts «خلايا بناء»

- الطبقة الداخلية تدي خلايا / الطبقة برا، جيبة تعمل حماية

No ....

→ There's 2 ways of Cartilage growth:-

a- interstitial growth:

As in cell nest - each cell 'll divide & go away from each other and so that the size of the Cartilage increase.

b- Appositional growth:-

differentiation into Chondrocytes by chondroblasts

• ال chondro blasts بيدي ال chondro cytes وتكبرها في جنب.

• Hyaline Cartilage found in respiratory Tree..

---

③ Elastic Cartilage:-

• More flexible → as ground substance Contain Elastic fibers.

• Stained by special stain (verhoeff's or Weigert's stain).

→ H & E Can't differentiate between Hyaline Cartilage + Elastic Cartilage As They Have The Same structure

But the reason of the difference is the Elastic fibers in Elastic Cartilage.

• Chondrocytes May be found in isogenous group (cell nest).

• Perichondrium is found.

• found in Ear Penna / "External Ear" + Epiglottis.

Date ___ / ___ / ___ No ______

• Bone :-

→ from Supportive Connective Tissue
→ Consist of cells.

① osteoprogenitor cell :-

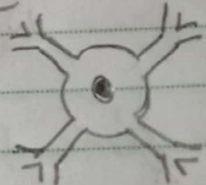
→ They are stem cells.
• خلايا جزعية لسة متخصصتش تقدر تدى اى حاجة ليها Processes.

② osteoblasts :-

→ The Mother cell of Bones
→ responsible for synthesis of bone Matrix (Collagen + glycoProtein)
→ found on Bone Surfaces side by side as epithelium شكله
→ it's Cuboidal in shape. + have cytoplasm Processes.
→ it's Cytoplasm is basophilic (Active Cells)
have rER + ribosomes ↑↑.
→ when it's Activity ↓ , the cell become flattened in shape
+ cytoplasmic basophilia ↓
→ اول ما تصنع بـ bone Matrix تتحول الى osteocyte

③ osteocyte :- ~~...~~ خلية الـ Bone

→ found in Lacuna + it's bone Matrix have organic & inorganic substances
← بـ Cartilage عكس organic بس
→ inorganic Sub. as K, Ca Make a gypusm around the cell
& it's isolated . & impermeable bone Matrix
ال Bone بيه خلية Blood vessels . انما ال Cartilage بيتغذى بالـ diffusion
• بيتواصل مع بعض بالـ ... جميعا يتقوا انت بسميها inter Cellular Canalicular



→ new osteocyte it's lacuna is spherical but the old one is Compressed.
→ The Cytoplasmic Processes are Called filo Podia.

Date / / No

4) osteoclasts :-

→ it's The Phagocyte cell of the bone. → Bone resorbing cell.
→ it's very large branched Motile cells. ← كبيرة أكولة متحركة متفرعة
→ The dilated Part of the cell body have from 2 : 50 nuclei (it's origin)
→ Howships lacunia is a shallow depression found in Bone Surface " due to bone resorbing "

- osteoclasts formed from osteoprogenator ✗
- osteoclasts formed from fusion of blood Monocyte ✓✓ → ←
- osteoplasts formed from osteoproginator cells ✓✓ → ←

→ The Cytoplasm of osteoclast is Acidophilic As it has Many Mitochondria + lysosomes.
→ found on surface of bones & have ruffled borders ᴖᴖᴖ (infolding of plasma Membrane).
→ it Produce [illegible] enzymes الحاجة آخرها ((ase))

⇒ Para thormone hormone ↑ activity of osteoclast.
⇒ Calcitonin hormone ↓ activity of osteoclast.

→ Bone Matrix :-
a- inorganic (Bone Minerals) as Ca, K, HCO_3, Mg
→ They form hydroxy apatite Crystals.

b- organic (osteoid) represent 50%
Composed of Collagen I + GAGS with Protein.

→ Bone Surface:

a- Periosteum:

- it's the ~~outer layer~~ External surface of bone.
- it Consists of outer layer of dense C.T. & the inner is Cellular.

⇒ it Makes fibers which Penetrates the bone to fix Periosteum in the bone & these fibers Called "Sharpey's fibers."

b- Endosteum:

- it's the inner Surface of bone
- Consists of 2 layers Can't be differentiated from each other.
- Have High % of reticular Connective Tissue which line Marrow Cavity.

→ Bone Types:

a- Primary bone: Spongy / Cancellous.

→ irregular Lamellae of Collagen forming branching.
مفيهوش اى حاجة ، والتفرعات بطريقة عشوائية

→ Has low % of Minerals, High % of Cells

b- Secondary bone: Compact - Cortical - Dense. "in adult"

→ it's Collagen fibers are arranged in Lamellae Concentrically around Haversians Canal. osteocytes ل خلـ

→ They are // to each other forming Haversian System.

ل Haverisan Canal , بيتصل Haversian Canal تانية كده تماما بالعرض
، بـــ Volkmann's Canal

⇒ Bone growth (interstitial) is Not found.
Appositional growth ✓✓✓

Date ..../..../.... No ....

Bone function.

1- Support. Skeleton.
2- Locomotion. Voluntary Muscles
3- Protectn Skull ⟶ Brain.
4- Manufacturing of blood cells (bone Marrow)
5- Homeo stasis. بيحافظ على الجسم